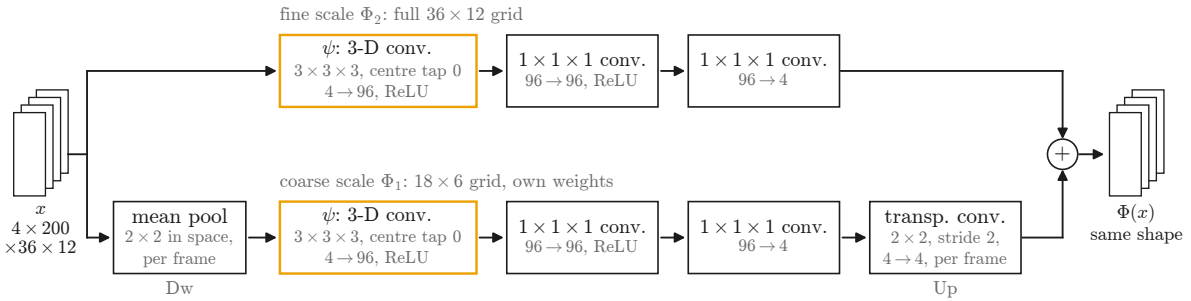
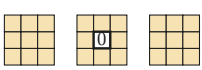
(a) prior $\Phi(x) = \mathrm{Up}(\Phi_1(\mathrm{Dw}(x))) + \Phi_2(x)$



(b) the kernel of ψ (one input/output channel pair)



The tap at the output's own point (t, i, j) is fixed to zero, and the layers after ψ act point by point, so a branch's output at s never reads $x(s)$. Pooling and the stride-2 transposed convolution act per frame, so each fine cell takes one coarse output, which excludes its own pooled cell. Hence $\Phi(x)(s)$ does not depend on $x(s)$ and Φ cannot learn the identity, which would make $\|x - \Phi(x)\|^2$ vanish.